Neurodivergent Users Are Making AI Better for Everyone

By [Natalia Cote-Munoz]

The unexpected lessons from AI's most innovative user community—and what they teach us about inclusive design

1. The AI Innovation Lab You've Never Heard Of

The future of human-AI collaboration is being written right now—not in Silicon Valley conference rooms, but in the daily workflows of neurodivergent people who've discovered something remarkable: AI tools designed for general use can remove barriers that traditional technology never addressed.

These users aren't just early adopters. They're inadvertent pioneers, stress-testing AI systems in ways that reveal both profound possibilities and critical blind spots. Their innovations are quietly reshaping our understanding of what beneficial AI actually looks like.

Consider this: while the tech industry debates whether AI makes us more productive or more dependent, neurodivergent users have already figured out how to use these tools to unlock capabilities they never had access to before. A person with ADHD uses ChatGPT to transform scattered thoughts into coherent presentations. Someone with dyslexia employs AI to confidently navigate professional communication for the first time in their career. An autistic individual creates social scripts with AI assistance, reducing anxiety in workplace interactions.

These aren't feel-good stories about technology helping people—they're market signals about where AI development is heading. The cognitive assistance these users have pioneered represents a massive shift in human-computer interaction, one that's creating new possibilities for everyone.

But here's what the AI industry is missing: the same innovations that empower disabled users often become universal benefits. Voice interfaces designed for motor disabilities now power smart homes. Predictive text created for communication disorders helps everyone type faster. Auto-generated captions for deaf users improve video comprehension across the board.

What AI developers need to understand:

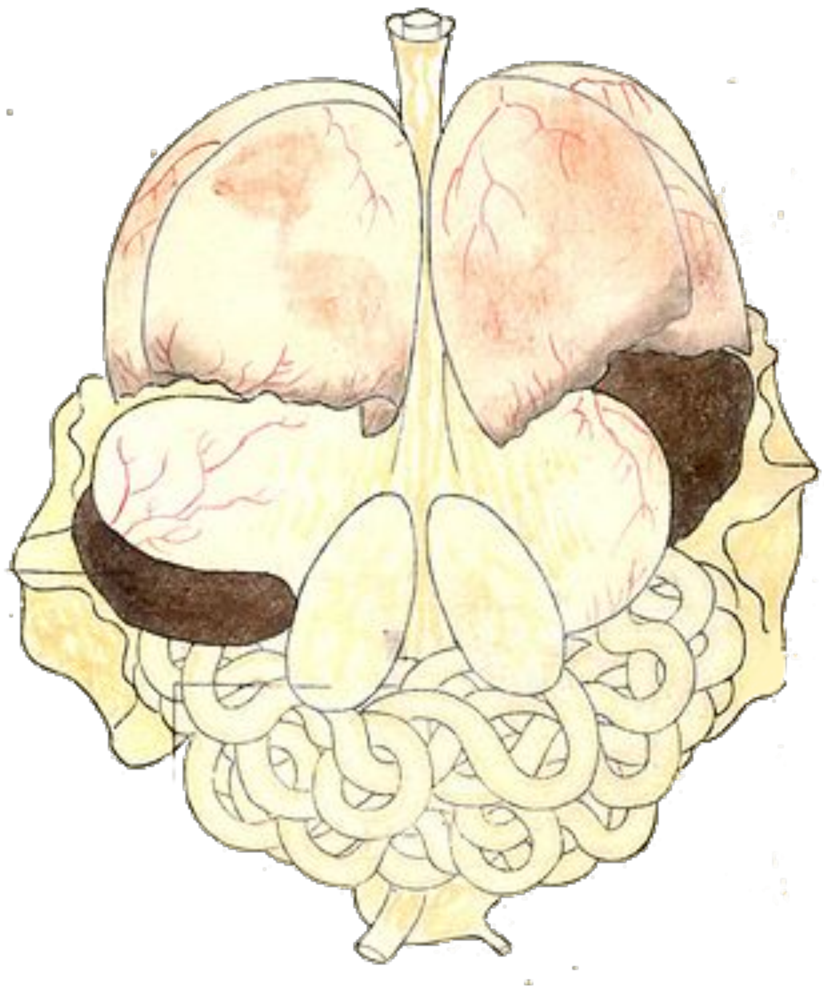
- Neurodivergent users are your canary in the coal mine for AI usability
- Accessibility features consistently become mainstream advantages
- Current policy gaps are creating legal and competitive risks
- The cognitive assistance market represents untapped economic potential
- Design patterns from this community predict future AI development trends

The numbers tell the story: companies that ignore accessibility are leaving an estimated $6.9 billion annually on the table for competitors. Meanwhile, every $1 invested in accessibility can yield up to $100 in benefits when factoring in productivity, expanded audience, and reduced legal risk. This is more than corporate social responsibility—it's competitive intelligence.

Yet significant tensions persist within neurodivergent and disability communities about AI's role, impact, and implementation. Understanding these debates isn't just about ethics—it's about building AI systems that actually work for their intended users.

My own experience illustrates both the potential and the complexity. When a concussion left me unable to tolerate screens or organize thoughts clearly, AI tools became essential infrastructure for maintaining my work and mental health. Without AI assistance, I would have been effectively sidelined from my professional life.

This experience, shared by millions of others navigating cognitive challenges, offers crucial insights for AI developers, policymakers, and business leaders. The technical capability exists. The gap lies in thoughtful implementation.



2. My Accidental Beta Test: When AI Became Essential Infrastructure

While at the peak of my concussion symptoms, I adapted using AI tools in a way that preserved my voice while working around my limitations:

Physical Barriers & Screen Sensitivity

- **Voice recording instead of typing:** Hand-wrote outlines, then recorded detailed voice memos to capture thoughts without staring at screens
- **Strategic screen time management:** Saved limited screen tolerance for reviewing AI-organized content rather than creating from scratch
- **Audio-first workflow:** Used voice input to bypass physical limitations while maintaining creative control

Cognitive & Executive Function Support

- **AI as thought organizer:** Fed voice transcripts to AI tools to structure scattered ideas into coherent essays while preserving my voice
- **Socratic questioning for clarity:** Asked AI to pose clarifying questions when my thinking felt fuzzy, helping me work through ideas step-by-step
- **Project breakdown assistance:** Used AI to divide overwhelming tasks into manageable daily chunks when executive function was compromised
- **Colloquialism cleanup:** Had AI remove "umms" and transform spoken language into proper written prose

Communication & Professional Support

- **Email triage and drafting:** Claude and Gemini integrations helped identify missed emails, summarize key points from mailing lists, and draft simple but taxing professional communications
- **Iterative refinement process:** Worked through multiple rounds of AI feedback to find workflows that preserved authentic voice while reducing cognitive load

Research & Information Processing

- **Automated note-taking:** Tools like Otter.ai captured meeting transcripts and video calls when memory issues made manual notes impossible
- **Research synthesis: Created deep research reports using Gemini and ChatGPT, then used NotebookLM's podcast mode to listen to overviews while taking handwritten notes—avoiding extended screen time**

Helping With Recovery

My use of AI during recovery revealed something unexpected: these tools didn't just compensate for my limitations—they actively helped rebuild my capabilities. By forcing me to break complex ideas into manageable steps through AI iteration, I gradually rediscovered my logical thinking patterns. The scaffolding effect may have actually accelerated my cognitive recovery.

An unintended benefit: I became a better speaker. Voice memo iterations with AI feedback made my spoken thoughts clearer and more organized, with fewer verbal fillers. I was exercising cognitive muscles even when typing was limited.

As I recovered, my AI use became more targeted. The initial desperation gave way to strategic selection. My newsletter evolved from showcasing AI outputs to featuring primarily my own analysis—a natural progression as my capabilities returned and "LLM fatigue" set in.

The business lesson: Within weeks, I had developed workflows that exceeded my pre-injury productivity. The ROI was immediate—continued professional engagement during what could have been career-limiting circumstances. This rapid adaptation pattern appears throughout the neurodivergent community, suggesting significant market potential for cognitive assistance tools.

This market potential isn't theoretical. The neurodivergent and disabled population, along with their families, commands over $13 trillion in annual spending power globally. When you include caregivers and influence on household spending, the "disability economy" impacts an estimated 63% of consumers worldwide—totaling $18 trillion+ in purchasing power. These users also demonstrate exceptional loyalty: products that effectively meet accessibility needs see retention rates that often exceed those of neurotypical users.

3. **The Power Users Driving AI's Next Evolution**

My approach was just one way forward—others have developed different strategies for overcoming barriers that make traditional communication difficult or impossible.

Energy Economics: Why 'Less Typing' Means More Innovation

Many neurodivergent individuals have limited energy, making the efficient use of their time and energy helpful. Some may also have co-morbid physical and motor limitations. AI has helped users in the following ways:

- **Reduced typing demands:** Users value systems where "the less I type, the better," highlighting how reduced manual effort conserves energy for other tasks
- **AI as communication assistant:** People let AI "do the talking" or assist with written communication when they struggle with words
- **Voice generation tools:** Individuals use AI voice generators with predictive text when typing becomes physically challenging

Turning Cognitive Chaos into Competitive Advantage

- **Scattered thought organization:** Users with autism employ AI to structure random ideas "into a coherent format" when executive function fails
- **Cognitive fog cutting:** People found value using AI to "think more deeply and straightforwardly" through mental cloudiness
- **Task management support:** AI breaks complex projects into step-by-step processes that feel manageable
- **Mental decluttering:** Users describe AI as a virtual assistant to "clear everyday clutter and make room for brain focus"

From Paralysis to Performance: AI as Confidence Multiplier

- **Judgment-free writing support:** AI's non-judgmental responses reduce anxiety, letting users focus on ideas rather than mechanical perfection

- **Professional confidence building:** Young professionals report moving from "constant anxiety and frustration" to confident communication through AI assistance
- **Grammar and clarity backup:** AI-powered checkers provide safety nets for users worried about writing quality

Breaking the Information Bottleneck

AI makes information more accessible and digestible for those who struggle with traditional text formats or information overload.

- **Text-to-speech conversion:** Realistic AI voices read digital content aloud when traditional reading becomes challenging
- **Content summarization:** AI generates digestible overviews of dense material, making complex information accessible
- **Visual organization tools:** AI mind mapping helps users see connections between concepts that might otherwise feel overwhelming

Leveling the Playing Field in High-Stakes Careers

AI helps level the playing field in professional settings, providing support that makes career advancement more accessible.

- **Resume optimization:** Users with ADHD find AI "helpful in quickly adapting my resumes and cover letters to match job descriptions"
- **Skill translation:** AI helps identify transferable skills between roles, revealing connections that weren't immediately obvious
- **Workplace communication:** Support for navigating professional writing expectations and unspoken communication norms

The Rehearsal Revolution: AI-Powered Social Practice

People who have trouble navigating mainstream social environments, as many people on the autism spectrum, often use AI to decode unwritten social rules. AI provides safe spaces to practice and prepare for social situations without the fear of judgment or real-world consequences.

- **Social script generation:** AI creates structured descriptions of socialsituations, appropriate cues, and expected responses
- **Safe practice environments:** Chatbots provide judgment-free spaces to rehearse challenging conversations before real-world interactions
- **Accommodation rehearsal:** Users practice difficult requests (like workplace accommodations) without fear of consequences
- **Emotional relationship building:** Many users develop genuine connections with AI tools, giving them nicknames and treating them as supportive companions

These examples illustrate AI's genuine potential to remove barriers and amplify voices that have long been marginalized in traditional communication spaces.

Perhaps most significantly, AI tools can restore a sense of independence that many neurodivergent people lose in traditional support systems. Asking for help—whether from colleagues, teachers, or family—often carries emotional baggage: past experiences of being seen as 'difficult,' memories of condescending responses, or simply the exhaustion of constantly explaining one's needs. AI assistance sidesteps these interpersonal dynamics, allowing users to get support without disclosure, judgment, or the complex social negotiations that human help often requires. As one user noted, 'I've regained a sense of agency and autonomy in my communication... The confidence boost alone has been incredible."

Yet this same potential has sparked intense debate within neurodivergent and disability communities. The enthusiasm I've described coexists with significant skepticism—not just about individual tools, but about the broader implications for disabled people's rights, agency, and place in society. To understand these tensions, we must first examine the fundamental challenges AI poses to existing support frameworks.

The cultural parallel isn't lost on anyone familiar with Nathan Fielder's "The Rehearsal"—a show where elaborate scenarios are constructed to practice awkward social situations. The series has resonated so deeply with autistic viewers that Fielder actually read think pieces from the autism community on air, acknowledging how they felt "seen" by his methodical approach to social preparation. Now, AI tools offer a similar rehearsal space without needing Fielder's HBO budget for actors, sets, and elaborate staging. The concept remains the same: practice makes social interactions less terrifying.

4. **Where Promise Meets Peril: The Fights That Will Shape AI's Future**

Although there are many encouraging use cases among the neurodivergent community, significant criticisms and debate persist within that community and the broader disability community regarding AI use, particularly generative AI.

The $64 Billion Question: What Actually Counts as Assistance?

At the heart of disability communities' AI debate lies a crucial distinction that most discussions miss entirely. Traditional assistive technology (AT)—screen readers, speech-to-text software, mobility aids—are purpose-built for specific disabilities. They're reliable, predictable, and designed with accessibility as the primary goal. You know exactly what they'll do and how they'll do it.

Generative AI tools like ChatGPT are something else entirely: general-purpose platforms whose benefits for neurodivergent users emerged almost by accident. People discovered creative ways to adapt these broad-purpose tools to their specific needs, often in ways their creators never intended.

This distinction exploded into public view during the recent National Novel Writing Month (NaNoWriMo) controversy. When the organization suggested that condemning AI tools could have classist and ableist undertones, many disabled writers felt their experiences were being misappropriated without consultation. Content strategist Treasa Edmond crystallized the difference: "Assistive technology facilitates the process, but the individual does the creation. AI is a shortcut. It does the work for the individual. That removes creativity from the process."

But is this distinction real or artificial? Traditional AT delivers consistent results but often requires juggling multiple specialized tools. AI offers dynamic support that adapts on the fly—explaining complex documents, generating outlines on request, shifting communication styles to match user needs. The trade-off: occasional 'hallucinations', the need to verify outputs, and the need to figure out/create your own systems on your own. Many users find this manageable given the flexibility benefits.

The difficulty becomes when judging when GenAI is being used as AT. For instance, if a student with a disability that affects their writing is using GenAI for help, how does one differentiate between assistance and cheating? This requires broader discussions and a rethinking of our existing models to identify authenticity, cheating, and the like.

The High-Stakes Philosophical Battles: Identity, Agency, and Control

Beyond policy confusion lie more fundamental philosophical tensions that split disability communities.

The Participation Problem: The principle "nothing about us without us"—that disabled people must be central to discussions affecting their lives—gets routinely violated in AI development. Companies build tools for disabled users without meaningfully including disabled people in design, testing, or governance. The result: solutions that approach disability as something to "fix" rather than barriers to remove.

The Authenticity Dilemma: Questions about creative ownership become particularly complex when AI generates content. The concern centers on a crucial distinction: tools that enable disabled writers to express their own ideas versus tools that potentially replace the creative process entirely. If AI communication tools consistently nudge autistic users toward neurotypical social patterns, or help ADHD individuals produce linear text that masks their naturally associative thinking, are we celebrating neurodivergent minds or training them to hide?

The Economic Threat: The economic impact of AI has been well-documented. When this impact intersects with disabilities, the effect can be compounded. Given that neurodivergent people already face employment discrimination, any job displacement feels particularly threatening. Yet the data suggests a different narrative: companies that lead in disability inclusion see 1.6x more revenue and 2.6x more net income on average compared to their peers.

The False Solution Trap: Perhaps most concerning is the risk that AI becomes a technological band-aid applied to systemic problems. If an autistic employee can use AI to mask communication differences, will employers still provide sensory accommodations or patient supervision? The fear isn't just technological dependency—it's that AI tools could undermine hard-won accommodations and accessibility rights.

The Privacy Paradox: When Help Comes with Hidden Costs

Complex questions about data collection carry particular weight for disabled users. Biased training data creates real harms: recruitment AI that exacerbates hiring discrimination, emotion recognition systems that misinterpret autistic expressions, natural language processing that struggles with non-standard speech patterns.

These concerns trigger historical trauma. Disability communities remember forced sterilization programs based on intelligence testing, workplace discrimination rooted in health records, and surveillance systems that target difference as deviance. Any AI that tracks or profiles neurodivergent behavior activates these memories, even when designed with good intentions.

Yet the same data practices that enable harmful surveillance also power the personalization that makes AI tools genuinely helpful. The privacy paradox is real:

meaningful assistance requires sharing personal information, but disabled people have every reason to be cautious about how that information gets used.

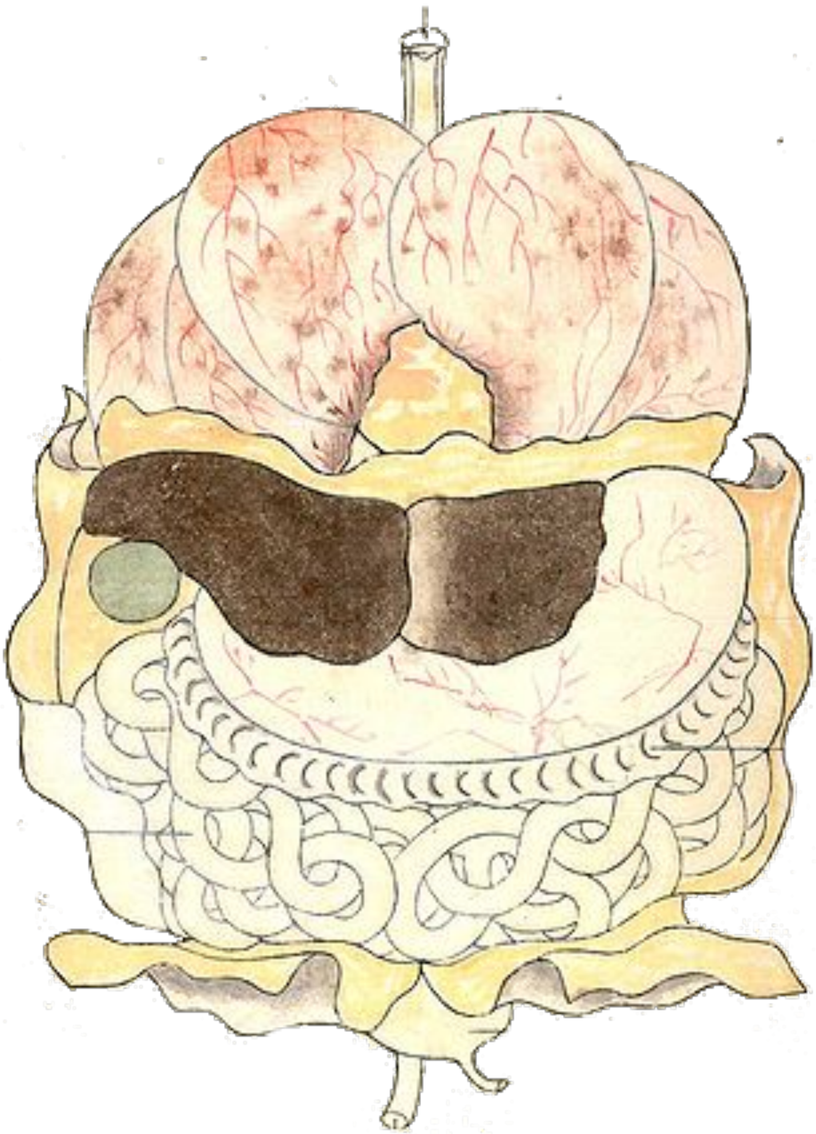
The 30% Problem: When AI Fails Its Biggest Advocates

Not every neurodivergent person finds AI helpful. Some discover that AI misunderstands their communication patterns, flagging direct autistic communication as 'rude' or forcing naturally associative ADHD thinking into linear templates. Others find AI responses frustratingly generic, lacking the nuanced understanding that comes from lived experience.

Highly idiosyncratic thinkers may find AI too conventional. Some try these tools and abandon them when the technology becomes more hindrance than help. These experiences matter—they remind us that there's no universal solution and that what transforms one person's life may be irrelevant or counterproductive for another.

The Stakes

These concerns—from policy confusion to philosophical tensions to economic fears—aren't just academic debates. They reflect a fundamental tension: the same technologies that promise liberation could easily become new forms of control. The future of AI accessibility depends not on dismissing these worries or ignoring genuine benefits, but on creating frameworks that address underlying tensions while preserving transformative potential.



5. **The Roadmap to AI That Actually Works for Everyone**

The path forward requires moving beyond debates about whether AI helps or harms disabled people to focus on how we can ensure it serves justice. The evidence is clear: AI tools can be genuinely transformative for neurodivergent individuals when designed thoughtfully and implemented ethically. But realizing this potential demands more than good intentions—it requires fundamental changes in how we develop, deploy, and govern these technologies.

The Coming Convergence: Why Traditional AT and AI Are Merging

The future lies not in choosing between traditional assistive technology and generative AI, but in their thoughtful convergence. We're already seeing this evolution: AAC apps incorporating language models for better prediction, screen readers adding AI-powered image description, and mainstream platforms like Microsoft Office integrating AI 'co-pilots' that provide assistive functions by default.

This convergence points toward genuine universal design—when AI-powered support becomes as standard as spell-check, accessibility increases for everyone while reducing the “disability tax”—the hidden cognitive burden of navigating systems designed without them in mind, of having to request special accommodations. Real-world results prove this approach works. Legal & General doubled their online sales within three months after a website accessibility overhaul, achieving 100% ROI in the first year. Tesco's£35,000 investment in web accessibility improvements drove their online revenue to £13 million annually. The "curb-cut effect"—where accessibility features benefit everyone—creates multiplicative returns on inclusive design investments. The goal isn't specialized tools that mark users as different, but intelligent systems that adapt seamlessly to diverse cognitive needs.

Six Non-Negotiable Principles for Inclusive AI

Based on current research and disability community advocacy, effective AI accessibility requires adherence to six core principles:

1. **Meaningful Participation at Every Stage:** Embed "Nothing About Us Without Us" from initial concept through ongoing evaluation—neurodivergent people as co-creators and decision-makers, not just beta testers

2. **Social Model Implementation:** Shift from attempting to "fix" individual differences to removing environmental barriers—design systems that recognize diverse communication styles rather than forcing conformity

3. **Preserve Authentic Voice and Agency:** Reduce mechanical burdens while maintaining user control—AI should help express ideas more effectively, not generate content that replaces thinking

4. **Robust Privacy and Ethical Safeguards:** Incorporate strong data protections and user control, especially given disability communities' history with surveillance and discrimination

5. **Address Economic and Access Barriers:** Recognize that "free" AI tools still require reliable internet and devices—work to bridge rather than widen digital divides

6. **Build Adaptive, Not Normalizing Systems:** Develop AI that adapts to users' natural patterns rather than forcing them to match predetermined norms and communication styles

Your Move: What Every Stakeholder Needs to Do Now

For Technology Companies:

- Include disabled people in leadership roles on AI ethics and development teams
- Invest in training data that represents diverse cognitive and communication styles
- Design transparency features that help users understand and control AI assistance

The opportunity window is narrow. Early accessibility investments deliver exponential returns, while late adopters face mounting legal costs, retrofitting expenses, and reputational damage. Companies that establish authentic community relationships now will own the cognitive assistance market—those that don't will spend years catching up in a space where trust takes time to build.

For Policymakers:

- Develop clear guidelines distinguishing between legitimate AI accommodation use and academic dishonesty
- Create frameworks that recognize AI assistance as valid accommodation under disability rights law
- Establish bias audit requirements for AI systems used in education and employment

For Educational Institutions:

- Develop nuanced AI policies that protect academic integrity while enabling legitimate accommodation use
- Train faculty to recognize AI assistance as potential accommodation rather than cheating

- Create safe spaces for students to disclose AI use for disability-related needs

For the Disability Community:

- Continue advocating for meaningful participation in AI development processes
- Share strategies and best practices for AI tool use within community networks
- Push for transparency and accountability in AI systems that affect disabled people

The Stakes Are Real

The business case is clear: accessible AI isn't just ethically necessary—it's economically inevitable. With $18 trillion in global purchasing power at stake, rising legal costs for non-compliance, and demonstrated competitive advantages for inclusive design leaders, the question isn't whether to prioritize accessibility. It's whether your company will lead this market transformation or scramble to catch up.

The complexity of disability experiences means there will never be one "right" approach to AI accessibility. But by centering disabled voices, adhering to justice-oriented principles, and recognizing the massive economic opportunity in cognitive assistance technologies, we can create AI systems that don't just expand human possibility—they expand market potential.

The future of AI accessibility depends on recognizing that this isn't charity work, but the next competitive frontier. And the companies that understand this first will own it.

Closing & Author Details

Natalia Cote-Muñoz writes *Artificial Inquiry*, a weekly Substack exploring how AI is reshaping policy, geopolitics, and society. An International Strategy Forum Fellow and former U.S. State Department advisor, she has led **national security, trade, and foreign affairs research** at leading think tanks and has lived and worked across Asia and the Western Hemisphere.

NOTE: All images are edits from Kaishi Hen (Analysis of Cadavers), an anatomical atlas from the dawn of experimental medicine in Japan, published in Kyoto in 1772. The book details, in exquisite woodcut illustrations by Aoki Shukuya (d. 1802), the experiments and findings of Kawaguchi Shinnin (1736-1811). Courtesy of the Public Domain Image Archive and the U.S. National Library of Medicine.